

In parallel, an electrical interface reads signals from neurons and delivers stimulation as needed. This setup involves electrodes and light-delivery systems that manage the electromagnetic radiation used to trigger molecular cages.

A robust software infrastructure supports these biological systems. A dedicated neural platform enables remote users to interact with living neurons by writing Python scripts. Custom interfaces are mainly developed in C++, with some tools in JavaScript. The entire system is open source, with documentation available on GitHub.

Equally important is how neurons handle time and energy. Unlike transistors, neurons remain mostly inactive, consuming little power. When triggered, a neuron sends a brief electrical spike lasting about 3 milliseconds—detectable as a voltage change. This sparse, event-driven behaviour contributes to the energy efficiency of neural systems.

This efficiency is the result of 300 million years of evolution. Neurons, which make up the nervous system, have been refined over time by natural selection to use power sparingly. Digital computers, in contrast, were built for speed and performance without the benefit of such long-term optimisation. Biological systems have evolved through random genetic variation to achieve efficiency levels that engineered systems are only beginning to approach.

Though neurons switch much more slowly, around 3 milliseconds compared to microseconds for transistors, this is not a limitation in practice. Digital systems can monitor neuron states and deliver stimulation at 30,000 cycles per second (30kHz), offering precise timing control for experiments.

Real-world applications are still about a decade away, but the foundation is in place. By combining biology, optics, electronics, and software, researchers are beginning

Market outlook and role in the future

Current demand. At present, global demand for biocomputers is close to zero. These systems are largely unknown to decision-makers, and practical applications are still emerging. Without real-world deployment, it is difficult to define the market accurately.

Forecasted growth. A recent study by Centric Business Intelligence estimates the current market value of biocomputers at \$7 billion, projected to grow to \$17 billion by 2032, with a CAGR of 13.13%. While the accuracy of these figures is uncertain, it is notable that such market projections have only started appearing this year, indicating growing interest in computational biology.

Silicon vs biocomputers. Biocomputers will not replace silicon-based systems for general tasks like cryptography or number crunching; such a replacement would be inefficient and impractical. However, for neuron simulations, biocomputers are expected to outperform digital systems in both energy efficiency and accuracy. Given that AI workloads are a major contributor to global electricity consumption, using biocomputers for neural simulations could offer significant energy savings.

to interface meaningfully with living computing systems.

How human skin cells are turned into brain-like structures

The stem cells used are called induced pluripotent stem cells (iPSCs). These are created by taking human skin cells and reprogramming them back into a stem-cell-like state. This technique, developed about 20 years ago, gives the cells pluripotency, meaning they can be turned into almost any other type of cell in the body.

Once the skin cells have been converted into iPSCs, researchers can guide them to become specific types of cells, such as neural stem cells. These, in turn, can be used to create organoids—the simplified, miniature versions of human organs. For example, a cortical organoid is not a full human cortex, but rather a small cluster of neurons with properties similar to the human cortex. These cells can even self-organise into structures that resemble parts of the brain, like the hippocampus or thalamus.

To keep these brain organoids alive and healthy, certain basic conditions must be maintained. First, they must be kept at a steady 37°C in an atmosphere containing 5% carbon dioxide and 80% humidity. The organoids are placed in a nutrient-rich fluid known as culture medium, which is mostly water mixed with amino acids, glucose, vitamins, and salts.

The medium needs to be

refreshed regularly, which is done using micropumps that circulate fresh medium around the organoids 24 hours a day. This ensures a stable environment where the cells can grow and function.

However, there are strict precautions to avoid what must not happen—especially contamination. The most dangerous type is bacterial contamination. If bacteria enter the medium, they can multiply rapidly—doubling every 20 minutes. Within 24 hours, the bacteria can consume all the glucose, create excess acid, and destroy the organoid.

Maintaining 5% CO₂ in the atmosphere helps keep the medium's pH at a neutral level of around 7. Without this, the medium would become too acidic, which would also harm or kill the neurons. This process of turning skin cells into brain-like tissue is delicate but powerful, offering new ways to study the brain and test treatments in the laboratory without using actual human brain tissue.

Electrodes that listen and speak

Tiny electrodes, just tens of microns wide, are placed in direct contact with brain organoids. These electrodes play a key role in both stimulating and recording from neurons. They can deliver very small electrical currents, typically around 1 microampere, to activate the neurons. They can also pick up tiny voltage changes, often as small as 100 microvolts, which are known as action potentials. These spikes